

9

The Schwarzschild geometry

We now consider how to solve the Einstein field equations and so discover the metric functions $g_{\mu\nu}$ in any given physical situation. Clearly, the high degree of *non-linearity* in the field equations means that a general solution for an arbitrary matter distribution is analytically intractable. The problem becomes easier if we look for special solutions, for example those representing spacetimes possessing *symmetries*. The first exact solution to Einstein's equations was found by Karl Schwarzschild in 1916.¹ As we shall see, the Schwarzschild solution represents the spacetime geometry outside a spherically symmetric matter distribution.

9.1 The general static isotropic metric

Schwarzschild sought the metric $g_{\mu\nu}$ representing the static spherically symmetric gravitational field in the empty space surrounding some massive spherical object such as a star. Thus, a good starting point for us is to construct the most general form of the metric for a static spatially isotropic spacetime.

A *static* spacetime is one for which some timelike coordinate x^0 (say) with the following properties: (i) all the metric components $g_{\mu\nu}$ are independent of x^0 ; and (ii) the line element ds^2 is invariant under the transformation $x^0 \rightarrow -x^0$. Note that (i) does not necessarily imply (ii), as is made clear by the example of a rotating star: time reversal changes the sense of rotation, but the metric components are constant in time. A spacetime that satisfies (i) but not (ii) is called *stationary*.

Thus, starting from the general expression for the line element

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu,$$

we wish to find a set of coordinates x^μ in which the $g_{\mu\nu}$ do not depend on the timelike coordinate x^0 and the line element ds^2 is invariant under $x^0 \rightarrow -x^0$, i.e.

¹ Astonishingly, Schwarzschild derived the solution while in the trenches on the Eastern Front during the First World War but sadly he did not survive the conflict.

the metric is *static*, and in which ds^2 depends only on *rotational invariants* of the spacelike coordinates x^i and their differentials, i.e. the metric is *isotropic*.

In fact, it is only slightly more complicated to derive the general form of the spatially isotropic metric *without* insisting that it is static. We therefore begin by constructing this more general metric. Only after its derivation will we impose the additional constraint that the metric is static.

The only rotational invariants of the spacelike coordinates x^i and their differentials are

$$\vec{x} \cdot \vec{x} \equiv r^2, \quad d\vec{x} \cdot d\vec{x}, \quad \vec{x} \cdot d\vec{x},$$

where $\vec{x} \equiv (x^1, x^2, x^3)$ and we have defined the coordinate r . Denoting the timelike coordinate x^0 by t , we thus find that the most general form of a spatially isotropic metric must be

$$ds^2 = A(t, r) dt^2 - B(t, r) dt \vec{x} \cdot d\vec{x} - C(t, r) (\vec{x} \cdot d\vec{x})^2 - D(t, r) d\vec{x}^2, \quad (9.1)$$

where A , B , C and D are *arbitrary* functions of the coordinates t and r .

Let us now transform to the (spherical polar) coordinates (t, r, θ, ϕ) , defined by

$$x^1 = r \sin \theta \cos \phi, \quad x^2 = r \sin \theta \sin \phi, \quad x^3 = r \cos \theta.$$

In this case, we have

$$\vec{x} \cdot \vec{x} = r^2, \quad \vec{x} \cdot d\vec{x} = r dr, \quad d\vec{x} \cdot d\vec{x} = dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2,$$

and so the general metric (9.1) now takes the form

$$ds^2 = A(t, r) dt^2 - B(t, r) r dt dr - C(t, r) r^2 dr^2 \\ - D(t, r) (dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2).$$

Collecting together terms and absorbing factors of r into our functions, thereby redefining A , B , C , D , the metric can be written

$$ds^2 = A(t, r) dt^2 - B(t, r) dt dr - C(t, r) dr^2 - D(t, r) (d\theta^2 + \sin^2 \theta d\phi^2).$$

If we now define a new radial coordinate by $\bar{r}^2 = D(t, r)$ and collect together terms into new arbitrary functions of t and $\bar{r}$, thereby again redefining A , B , C , we can write the metric as

$$ds^2 = A(t, \bar{r}) dt^2 - B(t, \bar{r}) dt d\bar{r} - C(t, \bar{r}) d\bar{r}^2 - \bar{r}^2 (d\theta^2 + \sin^2 \theta d\phi^2). \quad (9.2)$$

Let us also introduce a new timelike coordinate $\bar{t}$ defined by the relation

$$d\bar{t} = \Phi(t, \bar{r}) \left[A(t, \bar{r}) dt - \frac{1}{2} B(t, \bar{r}) d\bar{r} \right],$$

where $\Phi(t, r)$ is an integrating factor that makes the right-hand side an exact differential. Squaring, we obtain

$$d\bar{t}^2 = \Phi^2 \left(A^2 dt^2 - AB dt d\bar{r} + \frac{1}{4} B^2 d\bar{r}^2 \right),$$

from which we find

$$A dt^2 - B dt d\bar{r} = \frac{1}{A\Phi^2} d\bar{t}^2 - \frac{B}{4A} d\bar{r}^2.$$

Thus defining the new functions $\bar{A} = 1/(A\Phi^2)$ and $\bar{B} = C + B/(4A)$, our metric (9.2) becomes *diagonal* and takes the form

$$ds^2 = \bar{A}(\bar{t}, \bar{r}) d\bar{t}^2 - \bar{B}(\bar{t}, \bar{r}) d\bar{r}^2 - \bar{r}^2 (d\theta^2 + \sin^2 \theta d\phi^2).$$

There is no need to retain the bars on the variables, so we can write the metric as

$$ds^2 = A(t, r) dt^2 - B(t, r) dr^2 - r^2 (d\theta^2 + \sin^2 \theta d\phi^2). \quad (9.3)$$

Thus, the general isotropic metric is specified by *two* functions of t and r , namely $A(t, r)$ and $B(t, r)$. We will also see that, for surfaces given by t, r constant, the line element (9.3) describes the geometry of 2-spheres, which expresses the isotropy of the metric. In fact this line element shows that such a surface has a surface area $4\pi r^2$. However, because $B(t, r)$ is not necessarily equal to unity we cannot assume that r is the radial distance.

The final step in obtaining the most general *stationary* isotropic metric is now trivial. We require the metric functions $g_{\mu\nu}$ to be independent of the timelike coordinate, which means simply that A and B must be functions only of r . Thus, we have

$$ds^2 = A(r) dt^2 - B(r) dr^2 - r^2 (d\theta^2 + \sin^2 \theta d\phi^2). \quad (9.4)$$

Moreover, we see immediately that ds^2 is already invariant under $t \rightarrow -t$, and so this is the required form of the metric for a general static spatially isotropic spacetime.

9.2 Solution of the empty-space field equations

The functions $A(r)$ and $B(r)$ in the general static isotropic metric are determined by solving the Einstein field equations. We are interested in the spacetime geometry

outside a spherical mass distribution, so we must solve the empty-space field equations, which simply require the Ricci tensor to vanish:

$$R_{\mu\nu} = 0. \quad (9.5)$$

From equation (7.13) we can write the Ricci tensor as

$$R_{\mu\nu} = \partial_\nu \Gamma^\sigma_{\mu\sigma} - \partial_\sigma \Gamma^\sigma_{\mu\nu} + \Gamma^\rho_{\mu\sigma} \Gamma^\sigma_{\rho\nu} - \Gamma^\rho_{\mu\nu} \Gamma^\sigma_{\rho\sigma}, \quad (9.6)$$

and, in turn, the connection is defined in terms of the metric $g_{\mu\nu}$ by

$$\Gamma^\sigma_{\mu\nu} = \frac{1}{2} g^{\sigma\rho} (\partial_\nu g_{\rho\mu} + \partial_\mu g_{\rho\nu} - \partial_\rho g_{\mu\nu}). \quad (9.7)$$

Thus, we see that the deceptively simple expression (9.5) in fact equates to a rather complicated set of differential equations for the components of the metric $g_{\mu\nu}$.

To proceed further, we must calculate the connection coefficients $\Gamma^\sigma_{\mu\nu}$ corresponding to our static isotropic metric. This can be done in two ways. The quicker route (with any metric) is to use the Lagrangian procedure for geodesics discussed in Section 3.19. This involves writing down the ‘Lagrangian’

$$L = g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu,$$

in which $\dot{x}^\mu$ denotes $dx^\mu/d\sigma$, where σ is some affine parameter along the geodesic. Substituting L into the Euler–Lagrange equations then yields the equations of an affinely parameterised geodesic, from which the connection coefficients can be identified. Since later we will discuss the motion of particles in the Schwarzschild geometry, this procedure would be doubly beneficial.

For illustration, however, we will adopt the more traditional (but slower) method, in which the $\Gamma^\sigma_{\mu\nu}$ are calculated directly from the metric $g_{\mu\nu}$ using (9.7). Thus we first need to identify the metric components from the line element (9.4). The non-zero elements of $g_{\mu\nu}$ and $g^{\mu\nu}$ are

$$\begin{aligned} g_{00} &= A(r), & g^{00} &= 1/A(r), \\ g_{11} &= -B(r), & g^{11} &= -1/B(r), \\ g_{22} &= -r^2, & g^{22} &= -1/r^2, \\ g_{33} &= -r^2 \sin^2 \theta, & g^{33} &= -1/(r^2 \sin^2 \theta), \end{aligned}$$

where we note that the contravariant components of the metric are simply the reciprocals of the covariant components, since the metric is diagonal.

Substituting the metric components into the expression (9.7) for the connection, we find the expressions given in Table 9.1 (with no sums on Latin indices) with all the other components equalling zero. Thus, summarising these results, we find

Table 9.1 *The connection coefficients of the general static isotropic metric*

$\Gamma^0_{00} = 0$		
$\Gamma^i_{00} = -\frac{1}{2}g^{i\rho}\partial_\rho g_{00}$	$\Rightarrow$	$\Gamma^1_{00} = \frac{1}{2B(r)} \frac{dA(r)}{dr}$
$\Gamma^0_{0i} = \frac{1}{2}g^{0\rho}(\partial_i g_{\rho 0} + \partial_0 g_{\rho i} - \partial_\rho g_{0i}) = \frac{1}{2}g^{00}\partial_i g_{00}$	$\Rightarrow$	$\Gamma^0_{01} = \frac{1}{2A(r)} \frac{dA(r)}{dr}$
$\Gamma^0_{ij} = 0$		
$\Gamma^i_{ii} = \frac{1}{2}g^{i\rho}(\partial_i g_{\rho i} + \partial_i g_{\rho i} - \partial_\rho g_{ii}) = \frac{1}{2}g^{ii}\partial_i g_{ii}$	$\Rightarrow$	$\Gamma^1_{11} = \frac{1}{2B(r)} \frac{dB(r)}{dr}$
$\Gamma^1_{22} = \frac{1}{2}g^{11}(\partial_2 g_{12} + \partial_2 g_{12} - \partial_1 g_{22})$	$\Rightarrow$	$\Gamma^1_{22} = -\frac{r}{B(r)}$
$\Gamma^1_{33} = -\frac{1}{2}g^{11}\partial_1 g_{33}$	$\Rightarrow$	$\Gamma^1_{33} = -\frac{r \sin^2 \theta}{B(r)}$
$\Gamma^2_{21} = \frac{1}{2}g^{22}\partial_1 g_{22}$	$\Rightarrow$	$\Gamma^2_{21} = \frac{1}{r}$
$\Gamma^2_{33} = -\frac{1}{2}g^{22}\partial_2 g_{33}$	$\Rightarrow$	$\Gamma^2_{33} = -\sin \theta \cos \theta$
$\Gamma^3_{31} = \frac{1}{2}g^{33}\partial_1 g_{33}$	$\Rightarrow$	$\Gamma^3_{31} = \frac{1}{r}$
$\Gamma^3_{32} = \frac{1}{2}g^{33}\partial_2 g_{33}$	$\Rightarrow$	$\Gamma^3_{32} = \frac{\cos \theta}{\sin \theta}$

that only nine of the 40 independent connection coefficients are non-zero; they read as follows:

$$\begin{aligned}
 \Gamma^0_{01} &= A'/(2A), & \Gamma^1_{00} &= A'/(2B), & \Gamma^1_{11} &= B'/(2B), \\
 \Gamma^1_{22} &= -r/B, & \Gamma^1_{33} &= -(r \sin^2 \theta)/B, & \Gamma^2_{12} &= 1/r, \\
 \Gamma^2_{33} &= -\sin \theta \cos \theta, & \Gamma^3_{13} &= 1/r, & \Gamma^3_{23} &= \cot \theta.
 \end{aligned}$$

We now substitute these connection coefficients into the expression (9.6) in order to obtain the components $R_{\mu\nu}$ of the Ricci tensor. This requires quite a lot of tedious (but simple) algebra. Fortunately the off-diagonal components $R_{\mu\nu}$ for $\mu \neq \nu$ are identically zero, and we find that the diagonal components are

$$R_{00} = -\frac{A''}{2B} + \frac{A'}{4B} \left(\frac{A'}{A} + \frac{B'}{B} \right) - \frac{A'}{rB}, \quad (9.8)$$

$$R_{11} = \frac{A''}{2A} - \frac{A'}{4A} \left(\frac{A'}{A} + \frac{B'}{B} \right) - \frac{B'}{rB}, \quad (9.9)$$

$$R_{22} = \frac{1}{B} - 1 + \frac{r}{2B} \left(\frac{A'}{A} - \frac{B'}{B} \right), \quad (9.10)$$

$$R_{33} = R_{22} \sin^2 \theta. \quad (9.11)$$

The empty-space field equations (9.5) are thus obtained by setting each of the expressions (9.8–9.11) equal to zero. Of these four equations, only the first three are useful, since the fourth merely repeats the information contained in the third. Adding B/A times (9.8) to (9.9) and rearranging gives

$$A'B + AB' = 0,$$

which implies that $AB = \text{constant}$. Let us denote this constant by α . Substituting $B = \alpha/A$ into (9.10) we obtain $A + rA' = \alpha$, which can be written as

$$\frac{d(rA)}{dr} = \alpha.$$

Integrating this equation gives $rA = \alpha(r + k)$, where k is another integration constant. Thus the functions $A(r)$ and $B(r)$ are given by

$$A(r) = \alpha \left(1 + \frac{k}{r}\right) \quad \text{and} \quad B(r) = \left(1 + \frac{k}{r}\right)^{-1}.$$

In solving for A and B we have used only the sum of equations (9.8) and (9.9), not the separate equations. It is, however, straightforward to check that, with these forms for A and B , the equations (9.8–9.11) are satisfied separately.

It can be seen that the integration constant k must in some way represent the mass of the object producing the gravitational field, as follows. We can identify k (and α) by considering the weak-field limit, in which we require that

$$\frac{A(r)}{c^2} \rightarrow 1 + \frac{2\Phi}{c^2},$$

where Φ is the Newtonian gravitational potential. Moreover, in the weak-field limit r can be identified as the radial distance, to a very good approximation. For a spherically symmetric mass M we thus have $\Phi = -GM/r$, and so we conclude that $k = -2GM/c^2$ and $\alpha = c^2$. Therefore, the *Schwarzschild metric* for the empty spacetime outside a spherical body of mass M is²

$$ds^2 = c^2 \left(1 - \frac{2GM}{c^2 r}\right) dt^2 - \left(1 - \frac{2GM}{c^2 r}\right)^{-1} dr^2 - r^2 d\theta^2 - r^2 \sin^2 \theta d\phi^2.$$

(9.12)

We will use this metric to investigate the physics in the vicinity of a spherical object of mass M , in particular the trajectories of freely falling massive particles

² We note that the constant α could have been identified earlier by making the additional assumption that spacetime is *asymptotically flat*, i.e. that the line element (9.4) tends to the Minkowski line element in the limit $r \rightarrow \infty$. Thus we require that, in this limit, $A(r) \rightarrow c^2$ and $B(r) \rightarrow 1$ and so $AB = c^2$.

and photons. The Schwarzschild metric is valid down to the surface of the spherical object, at which point the empty-space field equations no longer hold. Clearly, the metric functions are infinite at $r = 2\mu$, which is known as the *Schwarzschild radius*. As we shall see, if the surface of the massive body contracts within this radius then the object becomes a *Schwarzschild black hole* (see Chapter 11). For the remainder of this chapter, however, we will restrict our attention to the region $r > 2GM/c^2$. We will often use the shorthand $\mu \equiv GM/c^2$ when writing down this metric.

9.3 Birkhoff's theorem

If we do not demand that our original metric is static (or stationary) but only that it is isotropic, then we would substitute the more general form (9.3),

$$ds^2 = A(t, r) dt^2 - B(t, r) dr^2 - r^2(d\theta^2 + \sin^2 \theta d\phi^2),$$

into Einstein's empty-space field equations $R_{\mu\nu} = 0$ in order to determine the functions $A(t, r)$ and $B(t, r)$. On repeating our earlier analysis, we would find some additional non-zero connection coefficients and components of the Ricci tensor. However, on solving this new set of equations, one discovers that the resulting metric must still be the Schwarzschild metric (9.12). Thus, we obtain *Birkhoff's theorem*, which states that *the spacetime geometry outside a general spherically symmetric matter distribution is the Schwarzschild geometry*.

This is an unexpected result because in Newtonian theory spherical symmetry has nothing to do with time dependence. This highlights the special character of the empty-space Einstein equations and of the solutions they admit. In particular, Birkhoff's theorem implies that if a spherically symmetric star undergoes strictly radial pulsations then *it cannot propagate any disturbance into the surrounding space*. Looking ahead to Chapter 18, this means that a radially pulsating star cannot emit gravitational waves.

One can show that the converse of Birkhoff's theorem is *not* true, i.e. a matter distribution that gives rise to the Schwarzschild geometry outside it *need not be spherically symmetric*. Indeed, some specific counter-examples are known.

9.4 Gravitational redshift for a fixed emitter and receiver

We begin our discussion of the physics in the vicinity of a spherical mass M by considering the phenomenon of *gravitational redshift*. In particular, we consider the specific example of an emitter, at *fixed* spatial coordinates (r_E, θ_E, ϕ_E) , which emits a photon that is received by an observer at *fixed* spatial coordinates (r_R, θ_R, ϕ_R) . If t_E is the coordinate time of emission and t_R the coordinate time

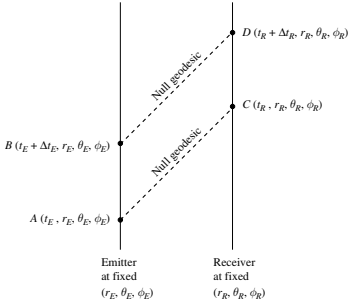


Figure 9.1 Schematic illustration of the emission and reception of two light signals.

of reception then the photon travels from the event $(t_E, r_E, \theta_E, \phi_E)$ to the event $(t_R, r_R, \theta_R, \phi_R)$ along a *null geodesic* in the Schwarzschild spacetime. This is illustrated schematically in Figure 9.1, which also shows a second photon, emitted at a later coordinate time $t_E + \Delta t_E$ and received at $t_R + \Delta t_R$.

In Appendix 9A we present an approach for calculating gravitational redshifts in more general situations. Nevertheless, in this simple case, it is instructive to derive the result in a more elementary manner: we need only use the fact that the photon geodesic is a null curve.³ Thus $ds^2 = 0$ at all points along it, and from the Schwarzschild metric (9.12), we find that

$$c^2 \left(1 - \frac{2\mu}{r}\right) dt^2 = \left(1 - \frac{2\mu}{r}\right)^{-1} dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2,$$

where we have written $\mu \equiv GM/c^2$. Let us consider the first signal. Thus, if σ is some affine parameter along the null geodesic then we have

$$\frac{dt}{d\sigma} = \frac{1}{c} \left(1 - \frac{2\mu}{r}\right)^{-1/2} \left[-g_{ij} \frac{dx^i}{d\sigma} \frac{dx^j}{d\sigma} \right]^{1/2},$$

³ This approach is based on that presented in J. Foster & J. D. Nightingale, *A Short Course in General Relativity*, Springer-Verlag, 1995.

where as before we use the notation $[x^\mu] = (t, r, \theta, \phi)$, the $g_{\mu\nu}$ are the components of the Schwarzschild metric and Latin indices run from 1 to 3. On integrating, we obtain

$$t_R - t_E = \frac{1}{c} \int_{\sigma_E}^{\sigma_R} \left(1 - \frac{2\mu}{r}\right)^{-1/2} \left[-g_{ij} \frac{dx^i}{d\sigma} \frac{dx^j}{d\sigma} \right]^{1/2} d\sigma,$$

where σ_E is the value of σ at emission and σ_R the value at reception. The important thing to notice about this expression is that the integral on the right-hand side depends only on the path through *space*. Thus, for a *spatially fixed* emitter and receiver, $t_R - t_E$ is the *same* for all signals sent. Thus the coordinate time difference Δt_E separating events A and B is equal to the coordinate time Δt_R between events C and D ,

$$\Delta t_R = \Delta t_E.$$

Now let us consider the *proper time* intervals along the worldlines of the emitter and receiver between each pair of events. Along both the emitter's and receiver's worldlines, $dr = d\theta = d\phi = 0$. Thus, from the Schwarzschild line element (9.12), in both cases

$$c^2 d\tau^2 \equiv ds^2 = c^2 \left(1 - \frac{2\mu}{r}\right) dt^2.$$

Moreover, in each case r is constant along the worldline, so we can immediately integrate this equation to obtain

$$\Delta\tau_E = \left(1 - \frac{2\mu}{r_E}\right)^{1/2} \Delta t_E \quad \text{and} \quad \Delta\tau_R = \left(1 - \frac{2\mu}{r_R}\right)^{1/2} \Delta t_R.$$

Thus, since $\Delta t_R = \Delta t_E$, we find that

$$\frac{\Delta\tau_R}{\Delta\tau_E} = \left(\frac{1 - 2\mu/r_R}{1 - 2\mu/r_E} \right)^{1/2},$$

which forms the basis of the formula for gravitational redshift. If we think of the two light signals as, for example, the two wavecrests of an electromagnetic wave, then it is clear that this ratio must also be the ratio of the *period* of the wave as observed by the receiver and emitter respectively. Thus the frequencies of the photon as measured by each observer are related by

$$\boxed{\frac{\nu_R}{\nu_E} = \left[\frac{1 - 2GM/(r_E c^2)}{1 - 2GM/(r_R c^2)} \right]^{1/2}}, \quad (9.13)$$

which shows that $\nu_R < \nu_E$ if $r_R > r_E$. The photon *redshift* z is defined by

$$1 + z = (\nu_R/\nu_E)^{-1}.$$

There is an important point to notice about this derivation. It can be generalised very easily to any spacetime in which we can choose coordinates such that the spacetime is stationary ($\partial_0 g_{\mu\nu} = 0$) and $g_{0i}(\vec{x}) = 0$. In this case,

$$ds^2 = g_{00}(\vec{x}) dt^2 + g_{ij}(\vec{x}) dx^i dx^j,$$

where, as indicated, all the metric components are *independent* of t . By repeating the above derivation for an emitter and observer at fixed spatial coordinates in this more general spacetime, we easily find that

$$\boxed{\frac{\nu_R}{\nu_E} = \left[\frac{g_{00}(\vec{x}_E)}{g_{00}(\vec{x}_R)} \right]^{1/2}}. \quad (9.14)$$

The derivations presented here depend crucially upon the fact that the emitter and receiver are spatially fixed. However, this is not often physically realistic. For example, we might want to calculate the gravitational redshift of a photon if the emitter or receiver (or both) are in free fall or moving in some arbitrary manner. A method for calculating redshifts in such general situations is given in Appendix 9A. In order to use this formalism, however, we require knowledge of the paths followed by freely falling particles and photons. Therefore, we now consider geodesics in the Schwarzschild geometry.

9.5 Geodesics in the Schwarzschild geometry

In deriving the Schwarzschild line element,

$$ds^2 = c^2 \left(1 - \frac{2\mu}{r} \right) dt^2 - \left(1 - \frac{2\mu}{r} \right)^{-1} dr^2 - r^2 d\theta^2 - r^2 \sin^2 \theta d\phi^2, \quad (9.15)$$

we also calculated the connection coefficients $\Gamma^\mu_{\nu\rho}$ for this metric. Thus we could now write down the geodesic equations for the Schwarzschild geometry in the form

$$\frac{d^2 x^\mu}{d\sigma^2} + \Gamma^\mu_{\nu\rho} \frac{dx^\nu}{d\sigma} \frac{dx^\rho}{d\sigma} = 0,$$

where σ is some affine parameter along the geodesic $x^\mu(\sigma)$. It is more instructive, however, to obtain the geodesic equations using the very neat ‘Lagrangian’ procedure discussed in Chapter 3.

Thus, let us consider the ‘Lagrangian’ $L = g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu$, where $\dot{x}^\mu \equiv dx^\mu/d\sigma$. Using (9.15), L is given by

$$L = c^2 \left(1 - \frac{2\mu}{r} \right) \dot{t}^2 - \left(1 - \frac{2\mu}{r} \right)^{-1} \dot{r}^2 - r^2 \left(\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2 \right). \quad (9.16)$$

The geodesic equations are then obtained by substituting this form for L into the the Euler–Lagrange equations

$$\frac{d}{d\sigma} \left(\frac{\partial L}{\partial \dot{x}^\mu} \right) - \frac{\partial L}{\partial x^\mu} = 0.$$

Performing this calculation, we find that the four resulting geodesic equations (for $\mu = 0, 1, 2, 3$) are given by

$$\left(1 - \frac{2\mu}{r}\right) \dot{t} = k, \quad (9.17)$$

$$\left(1 - \frac{2\mu}{r}\right)^{-1} \ddot{r} + \frac{\mu c^2}{r^2} \dot{t}^2 - \left(1 - \frac{2\mu}{r}\right)^{-2} \frac{\mu}{r^2} \dot{r}^2 - r \left(\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2 \right) = 0, \quad (9.18)$$

$$\ddot{\theta} + \frac{2}{r} \dot{r} \dot{\theta} - \sin \theta \cos \theta \dot{\phi}^2 = 0, \quad (9.19)$$

$$r^2 \sin^2 \theta \dot{\phi} = h. \quad (9.20)$$

In (9.17) and (9.20) respectively, the quantities k and h are *constants*. These two equations are derived immediately since L is not an explicit function of t or ϕ .

We see immediately that $\theta = \pi/2$ satisfies the third geodesic equation (9.19). Because of the spherical symmetry of the Schwarzschild metric we can therefore, with no loss of generality, confine our attention to particles moving in the ‘equatorial plane’ given by $\theta = \pi/2$. In this case our set of geodesic equations reduces to

$$\left(1 - \frac{2\mu}{r}\right) \dot{t} = k, \quad (9.21)$$

$$\left(1 - \frac{2\mu}{r}\right)^{-1} \ddot{r} + \frac{\mu c^2}{r^2} \dot{t}^2 - \left(1 - \frac{2\mu}{r}\right)^{-2} \frac{\mu}{r^2} \dot{r}^2 - r \dot{\phi}^2 = 0, \quad (9.22)$$

$$r^2 \dot{\phi} = h. \quad (9.23)$$

These equations are valid for both null and non-null affinely parameterised geodesics. In each of these cases, however, it is easier to replace the rather complicated r -equation (9.22) by a first integral of the geodesics equations. For a *non-null* geodesic this first integral is simply

$$g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu = c^2, \quad (9.24)$$

whereas for a *null* geodesic it is

$$g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu = 0. \quad (9.25)$$

Before going on to discuss separately non-null and null geodesics in the equatorial plane $\theta = \pi/2$, it is instructive to discuss the physical interpretation of the

constants k and h . One can arrive at equations (9.21) and (9.23) simply by using the fact that the components p_0 and p_3 of a particle's 4-momentum are conserved along geodesics since L does not depend explicitly on t and ϕ (remember that $\mathbf{p}$ is proportional to the tangent vector to the geodesic at each point). For notational simplicity, for a massive particle, we shall take the particle to have unit rest mass and choose the affine parameter to be the particle's proper time τ , so that $p^\mu = \dot{x}^\mu$. Similarly, for a massless particle we are free to choose an appropriate affine parameter along the null geodesic, once again such that $p^\mu = \dot{x}^\mu$. Thus, for $\theta = \pi/2$ we may write

$$p_0 = g_{00}\dot{t} = c^2 \left(1 - \frac{2\mu}{r}\right) \dot{t} = kc^2, \quad (9.26)$$

$$p_3 = g_{33}\dot{\phi} = -r^2\dot{\phi} = -h, \quad (9.27)$$

where in the last equality on each line we have defined the constants in a manner that coincides with (9.21) and (9.23). Let us first consider the constant k . If, at some event, an observer with 4-velocity $\mathbf{u}$ encountered a particle with 4-momentum $\mathbf{p}$ then he would measure the particle's energy to be

$$E = \mathbf{p} \cdot \mathbf{u} = p_\mu u^\mu.$$

For an observer at rest at infinity we have $[u^\mu] = (1, 0, 0, 0)$ and so $E = p_0 = kc^2$ (which is conserved along the particle geodesic). Thus we may take $k = E/c^2$, where E is the total energy of the particle in its orbit. Since for massive particles we have assumed unit mass, in the general case we have $k = E/(m_0 c^2)$, where m_0 is the rest mass of the particle. For the constant h , we can see immediately from (9.27) that it equals the specific angular momentum of the particle and that (as result of the choice of signature for the metric) p_3 is equal to *minus* the specific angular momentum. Finally, we note that the results (9.17–9.20) can also be derived using the alternative form (3.56) of the geodesic equations, which may be written

$$p_\mu = \frac{1}{2}(\partial_\mu g_{\nu\sigma})p^\nu p^\sigma. \quad (9.28)$$

9.6 Trajectories of massive particles

The trajectory of a massive particle is a timelike geodesic. Considering motion in the equatorial plane, we replace the geodesic equation (9.22) by (9.24), where $g_{\mu\nu}$ is taken from (9.15) with $\theta = \pi/2$. Moreover, since we are considering a timelike geodesic we can choose our affine parameter σ to be the *proper time* τ along the

path. Thus we find that the worldline $x^\mu(\tau)$ of a massive particle moving in the equatorial plane of the Schwarzschild geometry must satisfy the equations

$$\left(1 - \frac{2\mu}{r}\right) \dot{t} = k, \quad (9.29)$$

$$c^2 \left(1 - \frac{2\mu}{r}\right) \dot{t}^2 - \left(1 - \frac{2\mu}{r}\right)^{-1} \dot{r}^2 - r^2 \dot{\phi}^2 = c^2, \quad (9.30)$$

$$r^2 \dot{\phi} = h. \quad (9.31)$$

By substituting (9.29) and (9.31) into (9.30), we obtain the combined ‘energy’ equation for the r -coordinate,

$$\dot{r}^2 + \frac{h^2}{r^2} \left(1 - \frac{2GM}{c^2 r}\right) - \frac{2GM}{r} = c^2(k^2 - 1), \quad (9.32)$$

where we have written $\mu = GM/c^2$. We shall use this ‘energy’ equation to discuss radial free fall and the stability of orbits. Note that the right-hand side is a *constant* of the motion. We can verify the physical meaning of the constant k by noting that $E \propto k$. The constant of proportionality is fixed by requiring that, for a particle at rest at $r = \infty$, we have $E = m_0 c^2$. Letting $r \rightarrow \infty$ and $\dot{r} = 0$ in (9.32), we thus require $k^2 = 1$. Hence, as previously, we must have $k = E/(m_0 c^2)$, where E is the total energy of the particle in its orbit.

A second useful equation, which enables us to determine the *shape* of a particle orbit (i.e. r as a function of ϕ), may be found by using $h = r^2 \dot{\phi}$ to express $\dot{r}$ in the energy equation (9.32) as

$$\frac{dr}{d\tau} = \frac{dr}{d\phi} \frac{d\phi}{d\tau} = \frac{h}{r^2} \frac{dr}{d\phi}.$$

We thus obtain

$$\left(\frac{h}{r^2} \frac{dr}{d\phi}\right)^2 + \frac{h^2}{r^2} = c^2(k^2 - 1) + \frac{2GM}{r} + \frac{2GMh^2}{c^2 r^3}.$$

If we make the substitution $u \equiv 1/r$ that is usually employed in Newtonian orbit calculations, we find that

$$\left(\frac{du}{d\phi}\right)^2 + u^2 = \frac{c^2}{h^2}(k^2 - 1) + \frac{2GMu}{h^2} + \frac{2GMu^3}{c^2}.$$

We now differentiate this equation with respect to ϕ to obtain finally

$$\frac{d^2 u}{d\phi^2} + u = \frac{GM}{h^2} + \frac{3GM}{c^2} u^2. \quad (9.33)$$

In Newtonian gravity, the equations of motion of a particle of mass m in the equatorial plane $\theta = \pi/2$ may be determined from the Lagrangian

$$L = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\phi}^2) + \frac{GMm}{r}.$$

From the Euler–Lagrange equations we have

$$\begin{aligned} r^2\dot{\phi} &= h, \\ \ddot{r} &= \frac{h}{r^3} - \frac{GM}{r^2}, \end{aligned}$$

where the integration constant h is the specific angular momentum of the particle. If we now substitute $u = 1/r$ and eliminate the time variable, the Newtonian equation of motion for planetary orbits is obtained:

$$\frac{d^2u}{d\phi^2} + u = \frac{GM}{h^2}. \quad (9.34)$$

We must remember, however, that in this equation $u = 1/r$, where r is the radial *distance* from the mass, whereas in (9.33) r is a radial *coordinate* that is related to distance through the metric. Nevertheless, the forms of the two equations are very similar except for the extra term $3GMu^2/c^2$ in (9.33). We note that this term correctly tends to zero as $c \rightarrow \infty$.

Two interesting special cases of massive-particle orbits are worth investigating in detail, namely *radial motion* and *motion in a circle*.

9.7 Radial motion of massive particles

For radial motion ϕ is constant, which implies that $h = 0$. Thus, (9.32) reduces to

$$\dot{r}^2 = c^2(k^2 - 1) + \frac{2GM}{r}. \quad (9.35)$$

Differentiating this equation with respect to τ and dividing through by $\dot{r}$ gives

$$\ddot{r} = -\frac{GM}{r^2}, \quad (9.36)$$

which has precisely the same form as the corresponding equation of motion in Newtonian gravity. This does not imply, however, that general relativity and Newtonian gravity predict the same physical behaviour. It should be remembered that in (9.36) the coordinate r is not the radial distance, and dots indicate derivatives with respect to proper time rather than universal time.

As a specific example, consider a particle dropped from rest at $r = R$. From (9.35) we see immediately that $k^2 = 1 - 2GM/(c^2 R)$, so (9.35) can be written

$$\frac{\dot{r}^2}{2} = GM \left(\frac{1}{r} - \frac{1}{R} \right). \quad (9.37)$$

This has the same form as the Newtonian formula equating the gain in kinetic energy to the loss in gravitational potential energy for a particle (of unit mass) falling from rest at $r = R$. This provides a useful way to remember this equation, but the different meanings of r and of the dot should again be borne in mind.

We could continue our analysis of this quite general situation, but we can illustrate the main physical points by considering a particle dropped from rest at infinity. In this case $k = 1$ and the algebra is much less complicated. Thus, setting $k = 1$ in the geodesic equation (9.29) and in (9.35), we obtain

$$\frac{dt}{d\tau} = \left(1 - \frac{2\mu}{r} \right)^{-1}, \quad (9.38)$$

$$\frac{dr}{d\tau} = - \left(\frac{2\mu c^2}{r} \right)^{1/2}, \quad (9.39)$$

where in (9.39) we have taken the negative square root. These equations form the basis of our discussion of a radially *infalling* particle dropped from rest at infinity. From these equations we see immediately that the components of the 4-velocity of the particle in the (t, r, θ, ϕ) coordinate system are simply

$$[u^\mu] \equiv \left[\frac{dx^\mu}{d\tau} \right] = \left(\left(1 - \frac{2\mu}{r} \right)^{-1}, - \left(\frac{2\mu c^2}{r} \right)^{1/2}, 0, 0 \right).$$

Equation (9.39) determines the trajectory $r(\tau)$. On integrating (9.39) we immediately obtain

$$\tau = \frac{2}{3} \sqrt{\frac{r_0^3}{2\mu c^2}} - \frac{2}{3} \sqrt{\frac{r^3}{2\mu c^2}},$$

where we have written the integration constant in a form such that $\tau = 0$ at $r = r_0$. Thus τ is the proper time experienced by the particle in falling from $r = r_0$ to a coordinate radius r .

Instead of parameterising the worldline in terms of the proper time τ , we can alternatively describe the path as $r(t)$, thereby mapping out the trajectory of the particle in the (t, r) coordinate plane. This is easily achieved by writing

$$\frac{dr}{dt} = \frac{dr}{d\tau} \frac{d\tau}{dt} = - \left(\frac{2\mu c^2}{r} \right)^{1/2} \left(1 - \frac{2\mu}{r} \right). \quad (9.40)$$

On integrating, we find

$$t = \frac{2}{3} \left(\sqrt{\frac{r_0^3}{2\mu c^2}} - \sqrt{\frac{r^3}{2\mu c^2}} \right) + \frac{4\mu}{c} \left(\sqrt{\frac{r_0}{2\mu}} - \sqrt{\frac{r}{2\mu}} \right) \\ + \frac{2\mu}{c} \ln \left| \left(\frac{\sqrt{r/(2\mu)} + 1}{\sqrt{r/(2\mu)} - 1} \right) \left(\frac{\sqrt{r_0/(2\mu)} - 1}{\sqrt{r_0/(2\mu)} + 1} \right) \right|,$$

where the choice of the integration constant gives $t = 0$ at $r = r_0$.

In particular we note that

$$\tau \rightarrow \frac{2}{3} \sqrt{\frac{r_0^3}{2\mu c^2}} \quad \text{as } r \rightarrow 0, \\ t \rightarrow \infty \quad \text{as } r \rightarrow 2\mu.$$

Evidently, the particle takes a *finite* proper time to reach $r = 0$. When the worldline is expressed in the form $r(t)$, however, we see that r asymptotically approaches 2μ as $t \rightarrow \infty$. Since the coordinate time t corresponds to the proper time experienced by a stationary observer at large radius, we must therefore conclude that, to such an observer, it takes an *infinite* time for the particle to reach $r = 2\mu$. We return to this point later.

It is interesting to ask what velocity a stationary observer at r measures for the infalling particle as it passes. From the Schwarzschild metric (9.15) we see that, for a stationary observer at coordinate radius r , a coordinate time interval dt corresponds to a proper time interval

$$dt' = \left(1 - \frac{2\mu}{r} \right)^{1/2} dt.$$

Similarly, a radial coordinate separation dr corresponds to a proper radial distance measured by the observer equal to

$$dr' = \left(1 - \frac{2\mu}{r} \right)^{-1/2} dr.$$

Thus the velocity of the radially infalling particle, as measured by a stationary observer at r , is given by

$$\frac{dr'}{dt'} = \left(1 - \frac{2\mu}{r} \right)^{-1} \frac{dr}{dt} = - \left(\frac{2\mu c^2}{r} \right)^{1/2}. \quad (9.41)$$

Thus we find the rather surprising result that, as the particle approaches $r = 2\mu$, a stationary observer at that radius observes that the particle's velocity tends to c . We note that the equation (9.41) is only physically valid for $r > 2\mu$ since, as we shall see, it is impossible to have a stationary observer at $r \leq 2\mu$.

9.8 Circular motion of massive particles

For circular motion in the equatorial plane we have $r = \text{constant}$, and so $\dot{r} = \ddot{r} = 0$. Setting $u = 1/r = \text{constant}$ in the ‘shape’ equation (9.33) we have

$$u = \frac{GM}{h^2} + \frac{3GM}{c^2}u^2,$$

from which we find that

$$h^2 = \frac{\mu c^2 r^2}{r - 3\mu}.$$

Putting $\dot{r} = 0$ in the energy equation (9.32) and substituting the above expression for h^2 allows us to identify the constant k :

$$k = \frac{1 - 2\mu/r}{(1 - 3\mu/r)^{1/2}}. \quad (9.42)$$

The energy of a particle of rest mass m_0 in a circular of radius r is then given by $E = km_0c^2$. We can use this result to determine which circular orbits are *bound*. For this we require $E < m_0c^2$, so the limits on r for the orbit to be bound are given by $k = 1$. This yields

$$(1 - 2\mu/r)^2 = 1 - 3\mu/r,$$

which is satisfied when $r = 4\mu$ or $r = \infty$. Thus, over the range $4\mu < r < \infty$, circular orbits are bound. A plot of $E/(m_0c^2)$ as a function of r/μ is shown in Figure 9.2.

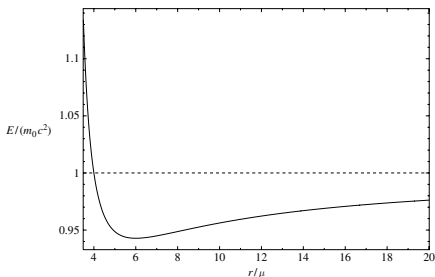


Figure 9.2 The variation of $k = E/(m_0c^2)$ as a function of r/μ for a circular orbit of a massive particle in the Schwarzschild geometry.

We can obtain another useful result by substituting our expression for h^2 into the geodesic equation $r^2 \dot{\phi} = h$; then we can write

$$\left(\frac{d\phi}{d\tau}\right)^2 = \frac{\mu c^2}{r^2(r-3\mu)}.$$

The significance of this equation is that it *cannot* be satisfied for circular orbits with $r < 3\mu$. Such orbits cannot be geodesics (since they do not satisfy the geodesic equations) and so cannot be followed by freely falling particles. Thus, according to general relativity a free massive particle cannot maintain a circular orbit with $r < 3\mu$ around a spherical massive body, no matter how large the angular momentum of the particle. This is very different from Newtonian theory.

It is also useful to calculate the expression for $d\phi/dt$, which is given by

$$\left(\frac{d\phi}{dt}\right)^2 = \left(\frac{d\phi}{d\tau} \frac{d\tau}{dt}\right)^2 = \frac{(1-2\mu/r)^2}{k^2} \left(\frac{d\phi}{d\tau}\right)^2 = \frac{\mu c^2}{r^3} = \frac{GM}{r^3}.$$

This expression is exactly the same as the Newtonian expression for the period of a circular orbit of radius r . Although we cannot say that r is the radius of the orbit in the relativistic case, we see that the spatial distance travelled in one complete revolution is $2\pi r$, just as in the Newtonian case.

9.9 Stability of massive particle orbits

The above analysis appears to suggest that the closest bound circular orbit around a massive spherical body is at $r = 4\mu$. However, we have not yet determined whether this orbit is stable.

In Newtonian dynamics the equation of motion of a particle in a central potential can be written

$$\frac{1}{2} \left(\frac{dr}{dt}\right)^2 + V_{\text{eff}}(r) = E,$$

where $V_{\text{eff}}(r)$ is the effective potential and E is the total energy of the particle per unit mass. For an orbit around a spherical mass M , the effective potential is

$$V_{\text{eff}}(r) = -\frac{GM}{r} + \frac{h^2}{2r^2}, \quad (9.43)$$

where h is the specific angular momentum of the particle. This effective potential is shown in Figure 9.3. It can be seen that bound orbits have two turning points and that a circular orbit corresponds to the special case where the particle sits at the minimum of the effective potential. Furthermore one sees that, in Newtonian dynamics, a finite angular momentum provides an *angular momentum barrier* preventing a particle reaching $r = 0$. This is *not* true in general relativity.

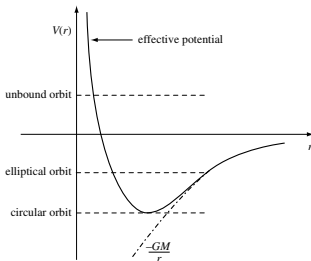


Figure 9.3 The Newtonian effective potential for $h \neq 0$, showing how an angular momentum barrier prevents particles reaching $r = 0$.

In general relativity, the ‘energy’ equation (9.32) for the motion of a particle around a central mass can be written

$$\frac{1}{2} \left(\frac{dr}{d\tau} \right)^2 + \frac{h^2}{2r^2} \left(1 - \frac{2\mu}{r} \right) - \frac{\mu c^2}{r} = \frac{c^2}{2} (k^2 - 1),$$

where we recall that the constant $k = E/(m_0 c^2)$. Thus in general relativity we identify the effective potential per unit mass as

$$V_{\text{eff}}(r) = -\frac{\mu c^2}{r} + \frac{h^2}{2r^2} - \frac{\mu h^2}{r^3}, \quad (9.44)$$

which has an additional term proportional to $1/r^3$ as compared with the Newtonian case (9.43). Remembering that $\mu = GM/c^2$, we see that (9.44) reduces to the form (9.43) in the non-relativistic limit $c \rightarrow \infty$.

Figure 9.4 shows the general relativistic effective potential for several values of $\bar{h} \equiv h/(c\mu)$. The dots indicate the locations of *stable* circular orbits, which occur at the local minimum of the potential. The local maxima in the potential curves are the locations of *unstable* circular orbits. For any given value of $\bar{h}$, circular orbits occur where $dV_{\text{eff}}/dr = 0$. Differentiating (9.44) gives

$$\frac{dV_{\text{eff}}}{dr} = \frac{\mu c^2}{r^2} - \frac{h^2}{r^3} + \frac{3\mu h^2}{r^4},$$

and so the extrema of the effective potential are located at the solutions of the quadratic equation

$$\mu c^2 r^2 - h^2 r + 3\mu h^2 = 0,$$

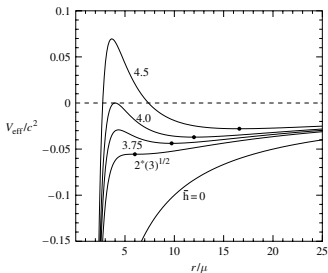


Figure 9.4 The general relativistic effective potential plotted for several values of the angular momentum parameter $\bar{h}$.

which occur at

$$r = \frac{h}{2\mu c^2} \left(h \pm \sqrt{h^2 - 12\mu^2 c^2} \right).$$

We note, in particular, that if $h = \sqrt{12}\mu c = 2\sqrt{3}\mu c$ then *there is only one extremum*, and there are no turning points in the orbit for lower values of h . The significance of this result is that the innermost stable circular orbit has

$$r_{\min} = 6\mu = \frac{6GM}{c^2}.$$

This orbit, with $r = 6\mu$ and $h/(\mu c) = 2\sqrt{3}$, is unique in satisfying both $dV_{\text{eff}}/dr = 0$ and $d^2V_{\text{eff}}/dr^2 = 0$, the latter being the condition for marginal stability of the orbit.

The existence of an innermost stable orbit has some interesting astrophysical consequences. Gas in an accretion disc around a massive compact central body settles into circular orbits around the compact object. However, the gas slowly loses angular momentum because of turbulent viscosity (the turbulence is thought to be generated by magnetohydrodynamic instabilities). As the gas loses angular momentum it moves slowly inwards, losing gravitational potential energy and heating up. Eventually it has lost enough angular momentum that it can no longer follow a stable circular orbit, and so it spirals rapidly inwards onto the central object.

We can make an estimate of the efficiency of energy radiation in an accretion disc. The maximum efficiency is of the order of the ‘gravitational binding energy’

at the innermost stable circular orbit (i.e. the energy E lost as the particle moves from infinity to the innermost orbit) divided by the rest mass energy of the particle. Setting $r = 6\mu$ in (9.42) and remembering that $k = E/(m_0c^2)$, we find that

$$\frac{E}{m_0c^2} = \frac{2\sqrt{2}}{3} \approx 0.943.$$

Thus the maximum radiation efficiency of the accretion disc is

$$\epsilon_{\text{acc}} \approx 1 - 0.943 = 5.7\%.$$

Thus, an accretion disc around a highly compact astrophysical object can convert perhaps a few percent of the rest mass energy of the gas into radiation; this may be compared with the efficiency of nuclear burning of hydrogen to helium (26 MeV per He nucleus),

$$\epsilon_{\text{nuclear}} \sim 0.7\%.$$

Accretion discs are therefore capable of converting rest mass energy into radiation with an efficiency that is about 10 times greater than the efficiency of the nuclear burning of hydrogen. The ‘accretion power’ of highly compact objects (such as black holes) cause some of the most energetic phenomena known in the universe.

A physically intuitive picture of a non-circular orbit and the capture of a particle with non-zero angular momentum h may be obtained by differentiating the energy

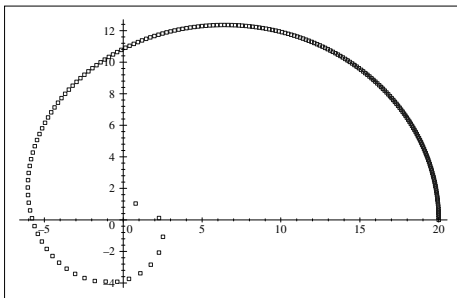


Figure 9.5 Orbit for a particle projected azimuthally from $r = 20GM/c^2$ with $h = 3.5GM/c$. A circular orbit would require $h = (20/\sqrt{17})GM/c$. The points are plotted at equal intervals of the particle's proper time.

equation (9.32) for massive particle orbits with respect to proper time τ . Using the original equation to remove the first derivative $dr/d\tau$, we find that

$$\frac{d^2 r}{d\tau^2} = -\frac{GM}{r^2} + \frac{h^2}{r^3} - \frac{3h^2 GM}{c^2 r^4}.$$

As we might expect, the first two terms on the right-hand side are very like the Newtonian expressions corresponding to an inward gravitational force and a repulsive ‘centrifugal force’ proportional to h^2 . The third term is new, however, and is also proportional to h^2 but this time acts *inwards*. This shows that close to a highly compact object, specifically within the radius $r = 3GM/c^2$, the centrifugal force ‘changes sign’ and is directed inwards, thus hastening the demise of any particle that strays too close to the object. This leads to spiral orbits of the type shown in Figure 9.5.

9.10 Trajectories of photons

The trajectory of a photon (and of any other particle having zero rest mass) is a null geodesic. We cannot use the proper time τ as a parameter, so instead we use some affine parameter σ along the geodesic. Considering motion in the equatorial plane, the equations of motion are given by the geodesic equations (9.21) and (9.23), and we replace the r -equation (9.22) by the condition $g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu = 0$. Thus we have

$$\left(1 - \frac{2\mu}{r}\right)\dot{t} = k, \quad (9.45)$$

$$c^2 \left(1 - \frac{2\mu}{r}\right)\dot{t}^2 - \left(1 - \frac{2\mu}{r}\right)^{-1}\dot{r}^2 - r^2\dot{\phi}^2 = 0, \quad (9.46)$$

$$r^2\dot{\phi} = h. \quad (9.47)$$

For photon trajectories, an analogue of the energy equation (9.32) can again be obtained by substituting (9.45) and (9.47) into (9.46), which gives

$$\boxed{\dot{r}^2 + \frac{h^2}{r^2} \left(1 - \frac{2\mu}{r}\right) = c^2 k^2.} \quad (9.48)$$

Similarly, the analogue for photons of the shape equation (9.33) is obtained by substituting $h = r^2\dot{\phi}$ into (9.46) and using the fact that

$$\frac{dr}{d\sigma} = \frac{dr}{d\phi} \frac{d\phi}{d\sigma} = \frac{h}{r^2} \frac{dr}{d\phi}.$$

Making the usual substitution $u \equiv 1/r$ and differentiating with respect to ϕ we find

$$\boxed{\frac{d^2 u}{d\phi^2} + u = \frac{3GM}{c^2} u^2.} \quad (9.49)$$

It is again worth mentioning the two special cases of radial motion and motion in a circle.

9.11 Radial motion of photons

For radial motion $\dot{\phi} = 0$ and (9.46) reduces to

$$c^2 \left(1 - \frac{2\mu}{r}\right) \dot{r}^2 - \left(1 - \frac{2\mu}{r}\right)^{-1} \dot{r}^2 = 0,$$

from which we obtain

$$\boxed{\frac{dr}{dt} = \pm c \left(1 - \frac{2\mu}{r}\right).} \quad (9.50)$$

On integrating, we have

$$ct = r + 2\mu \ln \left| \frac{r}{2\mu} - 1 \right| + \text{constant} \quad (\text{outgoing photon}),$$

$$ct = -r - 2\mu \ln \left| \frac{r}{2\mu} - 1 \right| + \text{constant} \quad (\text{incoming photon}).$$

Notice that under the transformation $t \rightarrow -t$, incoming and outgoing photon paths are interchanged, as we would expect. In fact for the moment the differential equation (9.50) is more useful. In a (ct, r) -diagram, we see that the photon worldlines will have slopes ± 1 as $r \rightarrow \infty$ (forming the standard special-relativistic lightcone), but their slopes approach $\pm \infty$ as $r \rightarrow 2\mu$. This means that they become more vertical; the cone ‘closes up’.

Our knowledge of the lightcone structure allows us to construct the ‘picture’ behind our earlier algebraic result that a particle takes infinite coordinate time to reach the horizon; this is illustrated in Figure 9.6. The curved solid line is the worldline of a massive particle dropped from rest by an observer fixed at $r = R$. Since massive particle worldlines are confined within the forward lightcone in any event, the closing up of the lightcones forces the worldlines of massive particles to become more vertical as $r \rightarrow 2\mu$. Thus, the particle ‘reaches’ $r = 2\mu$ only at $t = \infty$. Further, suppose that at some point along its trajectory the particle emits a radially outgoing photon in the direction of the observer. The tangent to

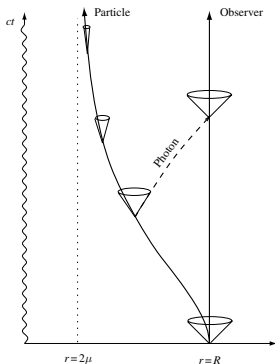


Figure 9.6 A radially infalling particle emitting a radially outgoing photon. The wavy line indicates the singularity at $r = 0$.

the resulting photon worldline must, at any event, lie along the outward-pointing forward lightcone at that point. This is illustrated by the broken line in Figure 9.6. Thus, in the limit where the particle approaches $r = 2\mu$, the initial direction of the photon worldline approaches the vertical and so the photon will be received by the observer only at $t = \infty$. Thus to an external observer the particle appears to take an infinite time to reach the horizon.

As discussed earlier, however, the *proper time* τ experienced by a massive particle in falling to $r = 2\mu$ is *finite*. Moreover, $dr/d\tau$ does not tend to zero at this point, so the particle has not ‘run out of steam’ and presumably passes beyond this threshold. Thus, our present coordinate system is inadequate for discussing what happens at and within $r = 2\mu$, and our (ct, r) -diagram is in some respects misleading in these regions. We discuss this further in Chapter 11.

9.12 Circular motion of photons

For motion in a circle we have $r = \text{constant}$. Thus, from the shape equation (9.49), we see that the only possible radius for a circular photon orbit is

$$r = \frac{3GM}{c^2}.$$

Therefore a massive object can have a considerable effect on the path of a photon. There is no such orbit around the Sun, for example, since the solar radius is much larger than $3GM_{\odot}/c^2 \approx 4.5$ km, but outside a black hole there can be such an orbit. As we shall see below, however, the orbit is not stable.

9.13 Stability of photon orbits

We can rewrite the ‘energy’ equation (9.48) for photon orbits as

$$\frac{\dot{r}^2}{h^2} + V_{\text{eff}}(r) = \frac{1}{b^2}, \quad (9.51)$$

where we have defined the quantity $b = h/(ck)$ and the effective potential

$$V_{\text{eff}}(r) = \frac{1}{r^2} \left(1 - \frac{2\mu}{r} \right).$$

In fact, by rescaling the affine parameter along the photon geodesic in such a way that $\lambda \rightarrow h\lambda$ the explicit h -dependence in (9.51) may be removed.

The effective potential is plotted in Figure 9.7, from which we see that $V_{\text{eff}}(r)$ has a single maximum at $r = 3\mu$, where the value of the potential is $1/(27\mu^2)$. Thus the circular orbit at $r = 3\mu$ is unstable. We conclude that there are no stable circular photon orbits in the Schwarzschild geometry.

The character of general photon orbits is determined by the value of the constant b . To find the physical meaning of b , we begin by using the geodesic equation (9.45) and the energy equation (9.47) to write

$$\frac{d\phi}{dr} = \frac{\dot{\phi}}{\dot{r}} = \frac{1}{r^2} \left[\frac{1}{b^2} - \frac{1}{r^2} \left(1 - \frac{2\mu}{r} \right) \right]^{-1/2}.$$

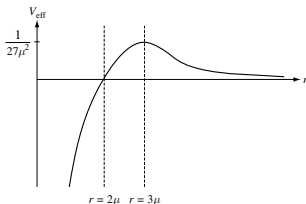


Figure 9.7 The effective potential for photon orbits.

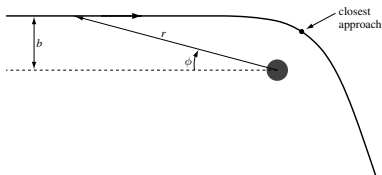


Figure 9.8 The shape of a photon orbit passing a spherical mass if $b > 3\sqrt{3}\mu$.

Thus, for a photon orbit, as $r \rightarrow \infty$ we have

$$r^2 \frac{d\phi}{dr} = \pm b.$$

Assuming that $\phi \rightarrow 0$ as $r \rightarrow \infty$, the solution to this equation is

$$r = \pm \frac{b}{\sin \phi},$$

which gives the equations of two straight lines with *impact parameter* b passing on either side of the origin.

The nature of the orbits depends very much on the value of the impact parameter b . Let us first consider inward-moving photons, i.e. photons for which r is initially decreasing. From (9.51) and Figure 9.7 we see that if $1/b^2 < 1/(27\mu^2)$, so that $b > 3\sqrt{3}\mu$, then the orbit will have a single turning point of closest approach and escape again to infinity. This situation is illustrated in Figure 9.8. If $b < 3\sqrt{3}\mu$, however, then the light ray will be captured by the massive body and spiral in towards the origin.

Similar considerations hold for trajectories that start at small radii. If $b > 3\sqrt{3}\mu$ then the photon will escape, and at infinity its straight-line path will have an impact parameter b . If $b < 3\sqrt{3}\mu$ then the photon path will have a turning point, and it will fall back towards the origin. In this case the particle does not reach infinity, so b cannot be interpreted simply as an impact parameter. It is straightforward to show that if a photon is emitted from within the region $r = 2\mu$ to $r = 3\mu$ then the opening angle α from the radial direction for the photon to escape varies from $\alpha = 0$ at $r = 2\mu$ to $\alpha = \pi/2$ at $r = 3\mu$.

Appendix 9A: General approach to gravitational redshifts

Consider a *general* spacetime with metric $g_{\mu\nu}$ in some arbitrary coordinate system x^μ , where x^0 is a timelike coordinate and the x^i are spacelike. Suppose that an

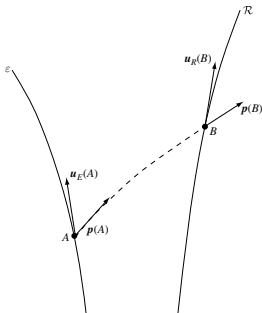


Figure 9.9 Schematic illustration of the emission and reception of a photon.

emitter $\mathcal{E}$ and a receiver $\mathcal{R}$ have worldlines $x_E^\mu(\tau_E)$ and $x_R^\mu(\tau_R)$ respectively, where τ_E and τ_R are the proper times of each observer. At some event A , $\mathcal{E}$ emits a photon with 4-momentum $\mathbf{p}(A)$ that is received by $\mathcal{R}$ at an event B . Furthermore, let us assume that at event A the emitter $\mathcal{E}$ has 4-velocity $\mathbf{u}_E(A)$ and that at event B the receiver has 4-velocity $\mathbf{u}_R(B)$. This is illustrated schematically in Figure 9.9.

The energies of the photon as observed by the emitter at A and by the receiver at B are respectively given by

$$E(A) = \mathbf{p}(A) \cdot \mathbf{u}_E(A) = p_\mu(A) u_E^\mu(A),$$

$$E(B) = \mathbf{p}(B) \cdot \mathbf{u}_R(B) = p_\mu(B) u_R^\mu(B).$$

Since in both cases $E = h\nu$, the ratio of the photon frequencies is given by the general result

$$\boxed{\frac{\nu_R}{\nu_E} = \frac{p_\mu(B) u_R^\mu(B)}{p_\mu(A) u_E^\mu(A)}}. \quad (9.52)$$

If we know the components of the 4-momentum $p_\mu(A)$ at emission then we can calculate the components $p_\mu(B)$ at reception, using the fact that the photon travels along a null geodesic. Since the photon 4-momentum $\mathbf{p}$ at any point is tangent to this geodesic, it is *parallel-transported* along the path. Thus, if the photon

geodesic $x^\mu(\sigma)$ is described in terms of some affine parameter σ then

$$\frac{dp_\mu}{d\sigma} - \Gamma^\nu_{\mu\rho} p_\nu \frac{dx^\rho}{d\sigma} = 0.$$

Moreover, since $\mathbf{p}$ is tangent to the geodesics, we can choose the affine parameter σ so that $p^\mu = dx^\mu/d\sigma$, in which case

$$\frac{dp_\mu}{d\sigma} = \Gamma^\nu_{\mu\rho} p_\nu p^\rho.$$

It is also worth remembering that a first integral of the equation for a photon geodesic, which can prove very useful, is

$$p_\mu p^\mu = 0. \quad (9.53)$$

Let us now examine some special cases of the general formula (9.52). We begin by considering the case in which both the emitter $\mathcal{E}$ and the receiver $\mathcal{R}$ have *fixed spatial coordinates*. Thus, for $i = 1, 2, 3$ the spatial components of their 4-velocities are

$$u_E^i \equiv \frac{dx_E^i}{d\tau_E} = 0 \quad \text{and} \quad u_R^i \equiv \frac{dx_R^i}{d\tau_R} = 0.$$

Moreover, in each case, the squared length of the 4-velocity is $g_{\mu\nu} u^\mu u^\nu = c^2$. In our situation, this reduces to $g_{00}(u^0)^2 = c^2$, so we find that

$$u^0 = \frac{c}{(g_{00})^{1/2}}.$$

Hence the formula (9.52) reduces to

$$\boxed{\frac{\nu_R}{\nu_E} = \frac{p_0(B)}{p_0(A)} \left[\frac{g_{00}(A)}{g_{00}(B)} \right]^{1/2}}. \quad (9.54)$$

Let us now make the *additional* assumption that the metric is *stationary* in our chosen coordinate system, i.e.

$$\partial_0 g_{\mu\nu} = 0.$$

Thus, the metric components $g_{\mu\nu}$ cannot depend explicitly on the coordinate x^0 . As shown in Section 3.19, this means that the zeroth covariant component of the tangent vector is *constant* along an affinely parameterised geodesic. Since the photon 4-momentum is simply proportional to the tangent vector, this means that

p_0 is constant along the photon's geodesic. Thus, in this case, (9.54) reduces further to

$$\frac{\nu_R}{\nu_E} = \left[\frac{g_{00}(A)}{g_{00}(B)} \right]^{1/2}, \quad (9.55)$$

and we have recovered the result (9.14) derived earlier.

Exercises

- 9.1 Show that surfaces of constant t and r in the general isotropic metric (9.3) have surface area $4\pi r^2$.
- 9.2 For the general static isotropic metric (9.4), show that the off-diagonal components of the Ricci tensor $R_{\mu\nu}$ are zero and that the diagonal components are given by (9.8–9.11).
- 9.3 The Schwarzschild line element is

$$ds^2 = c^2 \left(1 - \frac{2\mu}{r} \right) dt^2 - \left(1 - \frac{2\mu}{r} \right)^{-1} dr^2 - r^2 d\theta^2 - r^2 \sin^2 \theta d\phi^2.$$

By considering the 'Lagrangian' $L = g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu$, where the dots denote differentiation with respect to an affine parameter λ , calculate the connection coefficients $\Gamma^\mu_{\nu\sigma}$. Hence verify that the geodesic equations are given by (9.17–9.20).

- 9.4 Derive the results (9.17–9.20) using the alternative form (9.28) of the geodesic equations.
- 9.5 Calculate the connection coefficients and the Ricci tensor for the general isotropic metric (9.3). Hence prove Birkhoff's theorem.
- 9.6 Use Birkhoff's theorem to show that a particle inside a spherical shell of matter experiences no gravitational force.
- 9.7 Show that the 'Lemaître' line element

$$ds^2 = c^2 dw^2 - \frac{4}{9} \left[\frac{9\mu}{2(z-cw)} \right]^{2/3} dz^2 - \left[\frac{9\mu}{2(z-cw)} \right]^{2/3} d\Omega^2,$$

where $d\Omega^2 = d\theta^2 + \sin^2 \theta d\phi^2$, describes the Schwarzschild geometry. Show that observers with fixed spatial coordinates (z, θ, ϕ) are in free fall and had zero velocity at infinity, and that the proper time of such observers is w .

- 9.8 For a general stationary spacetime with line element

$$ds^2 = g_{00}(\vec{x}) dt^2 + g_{ij}(\vec{x}) dx^i dx^j,$$

show that, for a fixed emitter and receiver, the ratio of the received photon frequency to the emitted frequency is

$$\frac{\nu_R}{\nu_E} = \left[\frac{g_{00}(\vec{x}_E)}{g_{00}(\vec{x}_R)} \right]^{1/2},$$

where $\vec{x}_E$ and $\vec{x}_R$ are the fixed spatial coordinates of the emitter and receiver respectively.

- 9.9 An isolated thin rigid spherical shell has mass M and radius R . Suppose that a small hole is drilled through the shell, so that an observer O at the shell's centre can observe the outside universe. Show that a photon emitted by a fixed observer E at $r = r_E$ (where $r_E > R$) and received by O is blueshifted by the amount

$$\frac{\nu_O}{\nu_E} = \left(\frac{1 - 2\mu/r_E}{1 - 2\mu/R} \right)^{1/2}.$$

- 9.10 Show that the quantity

$$L^2 = p_\theta^2 + \frac{p_\phi^2}{\sin^2 \theta},$$

where $\mathbf{p}$ is the 4-momentum of a particle, is a constant of motion along any geodesic in the Schwarzschild geometry. Hence show that the particle orbits in a Schwarzschild geometry are stably planar.

- 9.11 For a particle dropped from rest at infinity in the Schwarzschild geometry, find expressions for $t(r)$ and $\tau(r)$, where t is the coordinate time and τ is the proper time of the particle.
- 9.12 A particle is dropped from rest at a coordinate radius $r = R$ in the Schwarzschild geometry. Obtain an expression for the 4-velocity of the particle in (t, r, θ, ϕ) coordinates when it passes coordinate radius r .
- 9.13 A particle at infinity in the Schwarzschild geometry is moving radially inwards with coordinate speed u_0 . Show that at any coordinate radius r the coordinate velocity is given by

$$\left(\frac{dr}{dt} \right)^2 = \left(1 - \frac{2GM}{c^2 r} \right)^2 c^2 \left[1 - \frac{1}{\gamma_0^2} \left(1 - \frac{2GM}{c^2 r} \right) \right],$$

where $\gamma_0 = (1 - u_0^2/c^2)^{-1/2}$. Determine the velocity relative to a stationary observer at r , and show that this velocity tends to c as r tends to $2GM/c^2$, irrespective of the value of u_0 .

- 9.14 Suppose that the particle in Exercise 9.13 has rest mass m_0 and that it stopped at $r = r_1$. If its excess energy was converted to radiation that is observed at infinity, show that the energy released as seen by a stationary observer at r_1 is

$$E = m_0 c^2 \left(\frac{\gamma_0}{\sqrt{1 - 2GM/(c^2 r_1)}} - 1 \right).$$

What is the energy released as observed at infinity? Show that this tends to $\gamma_0 m_0 c^2$ as r_1 tends to $2GM/c^2$.

- 9.15 For a particle in a circular orbit of radius r in the Schwarzschild geometry, use the alternative form (9.28) of the geodesic equations to show that

$$\frac{d\phi}{dt} = \frac{GM}{r^3}.$$

- 9.16 In the Schwarzschild geometry, a photon is emitted from a coordinate radius $r = r_2$ and travels radially inwards until it is reflected by a fixed mirror at $r = r_1$, so

that it travels radially outwards back to $r = r_2$. How long does the round trip take according to a stationary observer in infinity?

- 9.17 A photon moves in a circular orbit at $r = 3\mu$ in the Schwarzschild geometry. Show that the period of the orbit as measured by a stationary observer at this radius is $\tau = 6\pi\mu/c$. What is the period of the orbit as measured by a stationary observer in infinity?
- 9.18 Show that a massive particle moving in the innermost stable circular orbit in the Schwarzschild geometry has speed $c/2$ as measured by a stationary observer at this radius. Hence calculate the period of the orbit as measured by the local observer. What is the period of the orbit as measured by a stationary observer in infinity?
- 9.19 Alice is situated at a fixed position on the equator of the Earth (which is assumed to be spherical). In Schwarzschild coordinates (t, r, θ, ϕ) , her worldline is described in terms of a parameter τ by

$$t = \gamma\tau, \quad r = R, \quad \theta = \pi/2, \quad \phi = \omega\tau,$$

where γ and ω are constants and R is the coordinate radius of the Earth's surface. Bob is a distant stationary observer in space. Show that he will measure the orbital speed of Alice to be $v = R\omega/\gamma$. By considering the magnitude of Alice's 4-velocity, show that

$$\gamma = \left[1 - \left(\frac{v^2}{c^2} + \frac{GM}{c^2 r} \right) \right]^{-1/2},$$

where M is the mass of the Earth. Interpret this result physically.

- 9.20 All massive objects look larger than they really are. Show that a light ray grazing the surface of a massive sphere of coordinate radius $r > 3GM/c^2$ will arrive at infinity with impact parameter

$$b = r \left(\frac{r}{r - 2GM/c^2} \right)^{1/2}.$$

Hence show that the apparent diameter of the Sun ($M_\odot = 2 \times 10^{30}$ kg, $R_\odot = 7 \times 10^8$ m) exceeds the coordinate diameter by nearly 3 km.

- 9.21 The Hipparcos satellite can measure the positions of stars to an accuracy of 0.001 arcseconds. If it is measuring the position of a star in a direction perpendicular to the plane of the Earth's orbit, do Hipparcos observers need to account for the gravitational bending of light by the Sun?
- 9.22 A massive particle is moving in the equatorial plane of the Schwarzschild geometry. Show that at infinity the particle moves in a straight line with impact parameter $b = h/(c\sqrt{k^2 - 1})$.
- 9.23 An observer at rest at coordinate radius $r = R$ in the Schwarzschild geometry drops a massive particle which free-falls radially inwards. When the particle is at a coordinate radius $r = r_E$ it emits a photon radially outwards. Find an expression for the redshift z of the photon when it is received by the observer. Show that $z \rightarrow \infty$ as $r_E \rightarrow 2GM/c^2$.

- 9.24 Show that the geodesic equations for photon motion in the equatorial plane $\theta = \pi/2$ of the Schwarzschild geometry can be written in the form

$$\dot{r} = \frac{1}{bc} \left(1 - \frac{2\mu}{r}\right)^{-1}, \quad \dot{\phi} = \frac{1}{r^2}, \quad \dot{t}^2 = \frac{1}{b^2} - U(r),$$

where b is a constant, the dots correspond to differentiation with respect to some affine parameter and

$$U(r) = \frac{1}{r^2} \left(1 - \frac{2\mu}{r}\right).$$

Suppose that a photon moving in the equatorial plane passes an observer at rest at a coordinate radius in the range $2\mu < r < 3\mu$. Show that the observer measures the radial and azimuthal components of the photon's velocity to be

$$v_r = \pm c [1 - b^2 U(r)]^{1/2} \quad \text{and} \quad v_\phi = cb[U(r)]^{1/2}.$$

If the observer emits a photon that makes an angle α with the outward radial direction, show that the photon will escape to infinity provided that

$$\sin \alpha < 3\sqrt{3}\mu[U(r)]^{1/2}.$$

Find the values of α at $r = 2\mu$ and $r = 3\mu$.

- 9.25 Alice and Bob are astronauts in a space capsule, with no engine, in a circular orbit at $r = R$ (where $R > 3\mu$) in the equatorial plane of the Schwarzschild geometry. At some point in the orbit, Bob leaves the capsule, uses his rocket-pack to maintain a hovering position at that fixed point in space and then rejoins the capsule after it has completed one orbit. Show that the proper time interval measured by Alice while Bob is out of the capsule is

$$\Delta\tau_A = 2\pi \left[\frac{R^2}{\mu c^2} (R - 3\mu) \right]^{1/2}.$$

If $\Delta\tau_B$ is the corresponding proper time interval measured by Bob between these two events, show that

$$\frac{\Delta\tau_B}{\Delta\tau_A} = \left(\frac{R - 2\mu}{R - 3\mu} \right)^{1/2}.$$

Briefly compare this result with the 'twin paradox' result in special relativity. If Bob chooses not to rejoin the capsule but instead observes it fly past him, show that he will measure the capsule's speed as

$$v = \left(\frac{\mu c^2}{R - 2\mu} \right)^{1/2}.$$

- 9.26 A particle A and its antiparticle B are travelling in opposite senses in free circular orbits in the equatorial plane of the Schwarzschild geometry, one at coordinate

radius r_A and the other at $r_B (> r_A)$. At some instant, A emits a photon of frequency ν_A that travels radially and is received by B with frequency ν_B . Show that

$$\frac{\nu_B}{\nu_A} = \left(\frac{1 - 3\mu/r_A}{1 - 3\mu/r_B} \right)^{1/2}.$$

Suppose now that $r_A = r_B = r$, so that the particles collide and annihilate each other. Show that the total radiated energy as measured by an observer at rest at the point of collision is given by

$$E = 2m_0 c^2 \left(\frac{1 - 2\mu/r}{1 - 3\mu/r} \right)^{1/2},$$

where m_0 is the rest mass of each particle.

- 9.27 If the cosmological constant Λ is non-zero, show that the line element outside a static spherically symmetric matter distribution is given by

$$ds^2 = c^2 \left(1 - \frac{2\mu}{r} - \frac{\Lambda r^2}{3} \right) dt^2 - \left(1 - \frac{2\mu}{r} - \frac{\Lambda r^2}{3} \right)^{-1} dr^2 - r^2 (d\theta^2 + \sin^2 \theta d\phi^2).$$

Hence show that, in the weak-field Newtonian limit, a spherically symmetric mass M produces a gravitational field strength $\vec{g}$ given by

$$\vec{g} = \left(-\frac{GM}{r^2} + \frac{c^2 \Lambda r}{3} \right) \hat{r}.$$

Show that the shapes of massive particle orbits in the above geometry differ from those in the Schwarzschild geometry, but that the shapes of photon orbits do not.

- 9.28 Consider a static axisymmetric spacetime that is invariant under translations and reflections along the axis of symmetry. Show that, in general, the line element for such a spacetime can be written in the form

$$ds^2 = A(\rho) dt^2 - d\rho^2 - B(\rho) d\phi^2 - C(\rho) dz^2,$$

for arbitrary functions A , B and C . Show that the non-zero connection coefficients for this line element are given by

$$\begin{aligned} \Gamma^0_{01} &= \frac{A'}{2A}, & \Gamma^1_{22} &= -\frac{B'}{2}, & \Gamma^1_{33} &= -\frac{C'}{2}, \\ \Gamma^1_{00} &= \frac{A'}{2}, & \Gamma^2_{21} &= \frac{B'}{2B}, & \Gamma^3_{31} &= \frac{C'}{2C}, \end{aligned}$$

where the primes denote $d/d\rho$. Hence show that the non-zero components of the Ricci tensor are given by

$$\begin{aligned} R_{00} &= -\frac{A''}{2} + \frac{A'}{2} \left(\frac{A'}{2A} - \frac{B'}{2B} - \frac{C'}{2C} \right), \\ R_{11} &= \frac{A''}{2A} - \frac{(A')^2}{4A^2} + \frac{B''}{2B} - \frac{(B')^2}{4B^2} + \frac{C''}{2C} - \frac{(C')^2}{4C^2}, \\ R_{22} &= \frac{B''}{2} - \frac{B'}{2} \left(\frac{B'}{2B} - \frac{A'}{2A} - \frac{C'}{2C} \right), \\ R_{33} &= \frac{C''}{2} - \frac{C'}{2} \left(\frac{C'}{2C} - \frac{A'}{2A} - \frac{B'}{2B} \right). \end{aligned}$$

- 9.29 Consider a static, infinitely long, cylindrically symmetric matter distribution of constant radius that is invariant to Lorentz boosts along the symmetry axis (a 'cosmic string'). Show that the line element outside the body can be written as

$$ds^2 = c^2 dt^2 - d\rho^2 - (\alpha + \beta\rho)^2 d\phi^2 - dz^2,$$

where α and β are constants. For the case $\alpha = 0$, consider the spacelike surfaces defined by $t = \text{constant}$ and $z = \text{constant}$ and calculate the circumference of a circle of constant coordinate radius ρ in such a surface. Hence show that, for $\beta < 1$, the geometry on the spacelike surface is that of a two-dimensional cone embedded in three-dimensional Euclidean space.